

Chapter 2 Matrix Algebra

Section 2.1: Matrix Operations

Section 2.2: The Inverse of a Matrix

Section 2.3: Characterizations of Invertible Matrices

Matrix Operations

- If A is an $m \times n$ matrix—that is, a matrix with m rows and n columns—then the scalar entry in the i th row and j th column of A is denoted by a_{ij} and is called the (i, j) -entry of A .
- Each column of A is a list of m real numbers, which identifies a vector in $\mathbb{R}^m$.

The diagram illustrates the notation for a matrix A . It is represented as a large square bracket containing a grid of elements. The grid has m rows and n columns. The i th row is highlighted in light blue, and the j th column is also highlighted in light blue. The intersection of these row and column is the element a_{ij} . The elements in the i th row are $a_{i1}, \dots, a_{ij}, \dots, a_{in}$. The elements in the j th column are $a_{1j}, \dots, a_{ij}, \dots, a_{mj}$. The elements in the first, j th, and n th columns are labeled a_1, a_j, a_n respectively, with blue arrows pointing from these labels to the corresponding columns. The entire matrix is labeled $= A$ to the right of the bracket.

$$\begin{array}{c} \text{Column } j \\ \left[\begin{array}{ccccc} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ \text{Row } i & a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{array} \right] = A \\ \begin{array}{ccccc} \uparrow & & \uparrow & & \uparrow \\ \mathbf{a}_1 & & \mathbf{a}_j & & \mathbf{a}_n \end{array} \end{array}$$

Matrix notation.

Matrix Operations

- The columns are denoted by $\mathbf{a}_1, \dots, \mathbf{a}_n$, and the matrix A is written as $A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$
- The number a_{ij} is the i th entry (from the top) of the j th column vector $\mathbf{a}_j$.
- The **diagonal entries** in an $m \times n$ matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, a_{33}, \dots$, and they form the **main diagonal** of A .
- A **diagonal matrix** is an $n \times n$ matrix whose nondiagonal entries are zero.
- An example is the $n \times n$ identity matrix, I_n .

Sums and Scalar Multiples

- An $m \times n$ matrix whose entries are all zero is a **zero matrix** and is written as 0 .
- Two matrices are **equal** if they have the same size and their corresponding entries are equal.
- If A and B are $m \times n$ matrices, then the **sum** $A+B$ is the $m \times n$ matrix whose columns are the sums of the corresponding columns in A and B .

Sums and Scalar Multiples

The **sum** $A + B$ is defined only when A and B are the **same size**.

- **Example:** Let $A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix},$

and $C = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$. Find $A + B$ and $A + C$.

- **Solution:** $A + B = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix}$ but $A + C$ is not defined because A and C have different sizes.

Sums and Scalar Multiples

- If r is a scalar and A is a matrix, then the **scalar multiple** rA is the matrix whose columns are r times the corresponding columns in A .

Example: For $C = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$, compute $3C$.

Sums and Scalar Multiples

- **Theorem:** Let A , B , and C be matrices of the same size, and let r and s be scalars.
 - a. $A + B = B + A$
 - b. $(A + B) + C = A + (B + C)$
 - c. $A + 0 = A$
 - d. $r(A + B) = rA + rB$
 - e. $(r + s)A = rA + sA$
 - f. $r(sA) = (rs)A$
- Each quantity in Theorem 1 is verified by showing that the matrix on the left side has the same size as the matrix on the right and that corresponding columns are equal.

Matrix Multiplication

- **Row—column rule for computing AB**
- If a product AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B .
- If $(AB)_{ij}$ denotes the (i, j) -entry in AB , and if A is an $m \times n$ matrix, then

$$(AB)_{ij} = a_{i1}b_{1j} + \dots + a_{in}b_{nj}.$$

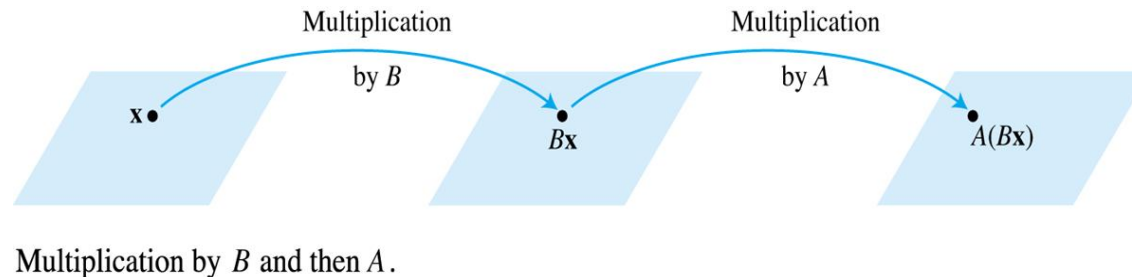
Matrix Multiplication

- **Definition:** If A is an $m \times n$ matrix, and if B is an $n \times p$ matrix with columns $\mathbf{b}_1, \dots, \mathbf{b}_p$, then the product AB is the $m \times p$ matrix whose columns are $A\mathbf{b}_1, \dots, A\mathbf{b}_p$.
- That is, $AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}$
- **Multiplication of matrices corresponds to composition of linear transformations.**

Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B .

Matrix Multiplication

- When a matrix B multiplies a vector $\mathbf{x}$, it transforms $\mathbf{x}$ into the vector $B\mathbf{x}$.
- If this vector is then multiplied in turn by a matrix A , the resulting vector is $A(B\mathbf{x})$.



- Thus $A(B\mathbf{x})$ is produced from $\mathbf{x}$ by a composition of mappings.

Matrix Multiplication

- **Example:** Compute AB , where

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}.$$

- **Solution:** Write $B = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \mathbf{b}_3]$ and compute :

Matrix Multiplication

$$A\mathbf{b}_1 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix},$$
$$= \begin{bmatrix} 11 \\ -1 \end{bmatrix}$$

$$A\mathbf{b}_2 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix},$$
$$= \begin{bmatrix} 0 \\ 13 \end{bmatrix}$$

$$A\mathbf{b}_3 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$
$$= \begin{bmatrix} 21 \\ -9 \end{bmatrix}$$

• Then

$$AB = A[\mathbf{b}_1 \quad \mathbf{b}_2 \quad \mathbf{b}_3] = \begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \end{bmatrix}$$

$\uparrow \quad \uparrow \quad \uparrow$
 $A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad A\mathbf{b}_3$

Properties of Matrix Multiplication

- **Theorem:** Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.
 - a. $A(BC) = (AB)C$ (associative law of multiplication)
 - b. $A(B + C) = AB + AC$ (left distributive law)
 - c. $(B + C)A = BA + CA$ (right distributive law)
 - d. $r(AB) = (rA)B = A(rB)$ for any scalar r
 - e. $I_m A = A = A I_n$ (identity for matrix multiplication)

Properties of Matrix Multiplication

- If $AB = BA$, we say that A and B **commute** with one another.
- **Warnings:**
 1. In general, $AB \neq BA$.
 2. The cancellation laws do **not** hold for matrix multiplication. That is, if $AB = AC$, then it is **not** true in general that $B = C$.
 3. If a product AB is the zero matrix, you **cannot** conclude in general that either $A = 0$ or $B = 0$.

Powers of a Matrix

- If A is an $n \times n$ matrix and if k is a positive integer, then A^k denotes the product of k copies of A :

$$A^k = \underbrace{A \cdots A}_k$$

- If A is nonzero and if $\mathbf{x}$ is in $\mathbb{R}^n$, then $A^k \mathbf{x}$ is the result of left-multiplying $\mathbf{x}$ by A repeatedly k times.
- If $k = 0$, then $A^0 \mathbf{x}$ should be $\mathbf{x}$ itself.
- Thus A^0 is interpreted as the identity matrix.

The Transpose of a Matrix

- Given an $m \times n$ matrix A , the **transpose** of A is the $n \times m$ matrix, denoted by A^T , whose columns are formed from the corresponding rows of A .

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

The Transpose of a Matrix

Theorem: Let A and B denote matrices whose sizes are appropriate for the following sums and products.

- a. $(A^T)^T = A$
- b. $(A + B)^T = A^T + B^T$
- c. For any scalar r , $(rA)^T = rA^T$
- d. $(AB)^T = B^T A^T$

The transpose of a product of matrices equals the product of their transposes in the **reverse** order.

Section 2.1: Matrix Operations

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Section 2.3: Characterizations of Invertible Matrices

Matrix Operations

- An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix C such that

$$CA = I \text{ and } AC = I$$

where $I = I_n$, the $n \times n$ identity matrix.

- In this case, C is an inverse of A .
- In fact, C is **uniquely determined** by A , because if B were another inverse of A , then

$$B = BI = B(AC) = (BA)C = IC = C$$

- This unique inverse is denoted by A^{-1} , so that

$$A^{-1}A = I \text{ and } AA^{-1} = I$$

Matrix Operations

- **Theorem:** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

- The quantity $ad - bc$ is called the determinant of A , and we write $\det A = ad - bc$
- This theorem says that a 2×2 matrix A is invertible if and only if $\det A \neq 0$.

Matrix Operations

- **Theorem:** If A is an invertible $n \times n$ matrix, then for each $\mathbf{b}$ in $\mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.
 - **Proof:** Take any $\mathbf{b}$ in $\mathbb{R}^n$.
 - A solution exists because if $A^{-1}\mathbf{b}$ is substituted for $\mathbf{x}$, then
$$A\mathbf{x} = A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = I\mathbf{b} = \mathbf{b}.$$
 - So $A^{-1}\mathbf{b}$ is a solution.
 - To prove that the solution is unique, show that if $\mathbf{u}$ is any solution, then $\mathbf{u}$ must be $A^{-1}\mathbf{b}$.
 - If $A\mathbf{u} = \mathbf{b}$, we can multiply both sides by A^{-1} and obtain
$$A^{-1}A\mathbf{u} = A^{-1}\mathbf{b}, I\mathbf{u} = A^{-1}\mathbf{b}, \text{ and } \mathbf{u} = A^{-1}\mathbf{b}.$$

Matrix Operations

- **Theorem:**

a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A$$

b. If A and B are $n \times n$ invertible matrices, then so is AB , and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}$$

c. If A is an invertible matrix, then so is A^T , and the inverse of A^T , is the transpose of A^{-1} . That is,

$$(A^T)^{-1} = (A^{-1})^T$$

Matrix Operations

- **Proof:** To verify statement (a), find a matrix C such that

$$A^{-1}C = I \text{ and } CA^{-1} = I$$

- These equations are satisfied with A in place of C . Hence A^{-1} is invertible, and A is its inverse.
- Next, to prove statement (b), compute:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

- A similar calculation shows that $(B^{-1}A^{-1})(AB) = I$.
- For statement (c), use Theorem 3(d), read from right to left,
 $(A^{-1})^T A^T = (AA^{-1})^T = I^T = I$.
- Similarly, $A^T (A^{-1})^T = I^T = I$.

Matrix Operations

- Hence A^T is invertible, and its inverse is $(A^{-1})^T$.
- The generalization of Theorem 6(b) is as follows:
The product of $n \times n$ invertible matrices is invertible, and the inverse is the product of their inverses in the reverse order.
- An invertible matrix A is row equivalent to an identity matrix, and we can find A^{-1} by watching the row reduction of A to I .

Elementary Matrices

- An **elementary matrix** is one that is obtained by performing a single elementary row operation on an identity matrix.

Elementary Matrices

- **Example:** Let $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}, A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

- Compute $E_1A, E_2A,$ and $E_3A,$ and describe how these products can be obtained by elementary row operations on A .

Elementary Matrices

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}, A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

- **Solution:** Verify that

$$E_1 A = \begin{bmatrix} a & b & c \\ d & e & f \\ g - 4a & h - 4b & i - 4c \end{bmatrix}, \quad E_2 A = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix},$$

$$E_3 A = \begin{bmatrix} a & b & c \\ d & e & f \\ 5g & 5h & 5i \end{bmatrix}.$$

- Addition of -4 times row 1 of A to row 3 produces $E_1 A$.

Elementary Matrices

- An interchange of rows 1 and 2 of A produces E_2A , and multiplication of row 3 of A by 5 produces E_3A .
- Left-multiplication by E_1 in Example 1 has the same effect on any $3 \times n$ matrix.
- Since $E_1I = E_1$, we see that E_1 itself is produced by this same row operation on the identity.

Elementary Matrices

- If an elementary row operation is performed on an $m \times n$ matrix A , the resulting matrix can be written as EA , where the $m \times m$ matrix E is created by performing the same row operation on I_m .
- Each elementary matrix E is invertible. The inverse of E is the elementary matrix of the same type that transforms E back into I .

Elementary Matrices

- **Theorem 7:** An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .
- **Proof:** Suppose that A is invertible.
- Then, since the equation $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b}$ (Theorem 5), A has a pivot position in every row.
- Because A is square, the n pivot positions must be on the diagonal, which implies that the reduced echelon form of A is I_n . That is, $A \sim I_n$.

Elementary Matrices

- Now suppose, conversely, that $A \sim I_n$.
- Then, since each step of the row reduction of A corresponds to left-multiplication by an elementary matrix, there exist elementary matrices $E_1, \dots, E_p$ such that

$$A \sim E_1 A \sim E_2 (E_1 A) \sim \dots \sim E_p (E_{p-1} \dots E_1 A) = I_n.$$

- That is, $E_p \dots E_1 A = I_n$ --- (1)
- Since the product $E_p \dots E_1$ of invertible matrices is

invertible, (1) leads to $(E_p \dots E_1)^{-1} (E_p \dots E_1) A = (E_p \dots E_1)^{-1} I_n$

$$A = (E_p \dots E_1)^{-1}$$

Algorithm for Finding Inverse of A

- Thus A is invertible, as it is the inverse of an invertible matrix (Theorem 6). Also,

$$A^{-1} = \left[(E_p \dots E_1)^{-1} \right]^{-1} = E_p \dots E_1.$$

- Then $A^{-1} = E_p \dots E_1 \cdot I_n$, which says that A^{-1} results from applying $E_1, \dots, E_p$ successively to I_n .
- This is the same sequence in (1) that reduced A to I_n .
- Row reduce the augmented matrix $[A \ I]$. If A is row equivalent to I , then $[A \ I]$ is row equivalent to $[I \ A^{-1}]$. Otherwise, A does not have an inverse.

Algorithm for Finding Inverse of A

- **Example:** Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}, \text{ if it exists.}$$

- **Solution:**

$$[A \quad I] = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix}$$

Algorithm for Finding Inverse of A

$$\sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix}$$

Reasonable Answers

- Theorem 7 shows, since $A \sim I$, that A is invertible, and

$$A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}.$$

- Now, check the final answer.

$$AA^{-1} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$